perplexity

Yeah its perfectly done, right now i m trying to make it work fo r v0.3.0 were we can have the same user for now (later shareable so ohter user) can revisit the trail.. i mnot sayign we need navigation but trace laong show the same trail different colors or somethign human can navigate across no need to .. maybe show the metric bu treal directions i m nto sure if u can do without ai then taht would be great too or via osm

Yes—v0.3.0 should introduce **Retrace**: return to a saved trail, see the original path alongside your current journey, and follow it yourself. It does not need to become a full navigation app.

For now, the same user revisits their own trail. Later, another person can follow the same saved trail through sharing.

And yes: basic route-following, progress, distances, and some direction cues can work without generative AI. The important distinction is between following an outdoor geographic trace and positioning yourself inside a previously recorded indoor space.

1. The v0.3.0 experience

The user opens a saved trail and chooses:

- Revisit — browse its recorded moments.
- Retrace — physically follow the journey again.

Retrace opens this experience:

```
Retracing: Evening courtyard walk

           [Map or aligned indoor plan]

           Original trail: dashed purple
           Current walk: solid teal
           You: location marker
           Moments: numbered markers

Next: Courtyard entrance
Approx. 45 m along the recorded trail
```

```
[Recenter]                  [View moments]

                  End retrace
```

Those distances are examples of interface content, not predetermined values.

The user interprets the path. Trails supplies context—not unsupported claims about which doorway is accessible or whether a path is currently safe.

2. Two paths, not one overwritten recording

Keep the original trail unchanged.

Create a separate retrace session containing:

```
RetraceSession
  id
  sourceTrailId
  startedAt
  endedAt?
  currentPath
  visitedMomentIds
  positioningMode
  alignmentState
```

Visual language

Element	Proposed treatment
Original recorded trail	Dashed purple
Current retrace path	Solid teal
Current position	Distinct position marker
Location uncertainty	Accuracy circle
Selected moment	Highlighted marker and card
Tracking gap	Broken line, not an invented connection

Use line styles and labels as well as colour.

The saved trail is the reference. The new path is what actually happened during this visit.

3. Outdoor retracing can be the dependable first release

For geographically recorded trails, the proposed implementation is:

1. Load the original coordinates.
2. Show the user’s current location.
3. Record their new path.

4. Estimate their position along the original trace.
5. Show relevant moments and approximate progress.
6. Preserve uncertainty rather than hiding it.

Core Location supplies geographic accuracy information; its horizontal-accuracy value represents a radius of uncertainty, not a guarantee that the displayed point is exact. This should directly affect the confidence of distance and deviation feedback. [1]

Useful metrics

- Distance walked during this retrace.
- Estimated distance remaining along the original trace.
- Approximate distance from the recorded path.
- Next moment.
- Estimated progress along the original trail.

Do not calculate progress from the geographically nearest point alone. On loops, crossings, or parallel paths, that can select the wrong part of the journey. The implementation should also consider recent progress, movement direction, and whether the match is ambiguous.

Honest status language

Prefer:

> “Near the recorded trail.”

> “Location accuracy is low.”

> “You appear to be away from the recorded path.”

Avoid:

> “Wrong way!”

when the positioning is not reliable enough to justify it.

4. Indoor retracing needs an explicit alignment step

This is the important limitation: displaying an old room model is not the same as knowing where the returning person is inside it.

For automatic indoor positioning, the new tracking session needs to align with the original session’s coordinate frame.

ARKit supports saving a world map and attempting to relocalise against it later. When relocalisation succeeds, the restored session can use the recorded spatial frame and anchors. It can also fail to reconcile the current environment with the saved map; success must not be assumed. [2] [3]

Proposed indoor flow

```
Open saved trail
→ Retrace indoors
→ Look around a recognisable starting area
→ Attempt alignment
→ Confirm aligned
→ Show original path and current local path
```

During alignment:

> "Look around the area where this trail began."

Only after successful alignment should Trails show a live position on the original indoor path.

If alignment fails

Offer:

- Retry alignment.
- Browse the saved plan and moments.
- Follow the sequence manually.
- Exit retrace.

Do not draw a moving marker on the original plan unless it is genuinely aligned.

A manual "I'm at this moment" action can support self-guided following, but it must not be presented as automatic localisation.

Implication for existing recordings

For future indoor recordings, save the relocalisation data alongside the local path where supported. Apple specifically recommends checking world-mapping quality before saving a world map for later use.[4]

Existing trails without suitable alignment data can still support manual revisiting; they should not automatically receive a "live indoor retrace" capability.

RoomPlan remains optional. The requirement is a usable shared coordinate frame—not simply the existence of a room model.

5. What directions can work without AI?

There are three different capabilities:

Capability	Generative AI needed?	Recommendation
Show original/current paths	No	Core v0.3.0
Estimate progress and distance along a trace	No	Core, with confidence handling

Capability	Generative AI needed?	Recommendation
Show the next recorded moment	No	Core
Generate simple bend cues from path geometry	No	Optional; noisy traces can produce misleading cues
Calculate street-network directions	No	Separate routing integration
Automatically know indoor position from an old model	Not solved by adding a chatbot	Requires positioning/alignment

A basic cue such as “The recorded trail bends left ahead” is different from “Take the next accessible doorway on your left.” The second requires knowledge the recorded line may not contain.

For this release, I would prioritise visual following and next-moment context over spoken turn instructions.

6. Where OpenStreetMap fits

OSM-based routing can help with a separate question:

> “How do I get from here to the start of this trail?”

For example, OSRM provides routing and a map-matching service that matches GPS traces to a road network. That is a different operation from faithfully displaying the original recording. [5]

Keep these actions distinct

- Get to start — a newly calculated route.
- Follow original trail — the saved recorded path.
- My current walk — the user’s new observations.

Do not silently replace the original trail with a road-network-matched route. The original journey may include areas that are not represented by that network.

An OSM routing integration also does not replace the indoor alignment requirement.

7. Recommended v0.3.0 scope

Release-blocking

- Retrace entry point on saved trails.
- Original and current paths shown distinctly.
- Current position and visible uncertainty.
- Recenter and fit-both-paths controls.
- Moment markers and manually selectable moment cards.
- Approximate progress when a reliable match exists.

- Separate retrace-session persistence.
- End and reopen a retrace session.
- Clear unsupported, low-accuracy, and alignment states.
- No changes to the original trail.

Conditional indoor capability

Enable live indoor retracing only for trails that have the required saved positioning data and successfully relocalise.

For all others, provide a clearly labelled manual follow-along experience.

Defer

- Automatic rerouting.
- Spoken turn-by-turn navigation.
- Guaranteed indoor positioning.
- Public sharing.
- Automatic floor detection.
- AI-generated directions.

8. The release test

The defining demonstration should be:

1. Record a real trail.
2. Finish and save it.
3. Return later.
4. Open Retrace.
5. See the old path and the new walk together.
6. Recognise and open the original moments.
7. Finish without overwriting the original.
8. Review both visits.

For indoor trails, add a separate test: live positioning appears only after successful alignment, and failure produces an honest fallback.

That gives v0.3.0 a clear identity: not “Trails now does navigation,” but “Trails lets you return and follow an experience again.”

⁂

1. https://developer.apple.com/documentation/corelocation/cllocation/horizontalaccuracy
2. https://developer.apple.com/documentation/arkit/saving-and-loading-world-data

3. https://developer.apple.com/documentation/arkit/arworldtrackingconfiguration/initialworldmap
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